EXPERIMENT NO. 1

Study Java Architecture and Develop a Basic Java Program

PART B – TO BE COMPLETED BY STUDENTS

Roll No. C60

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Class / Division SE-C

Batch C2

Date of Experiment _____

Date of Submission _____

Grade _____

B.1 Document Created by the Student

Create a Java source file, compile it, execute it, and paste the source code and output in the space provided.

Execution command: java PatientWelcome

Compilation command: javac PatientWelcome.java

Source file: PatientWelcome.java

B.2 Student Program / Output

Paste/write the final program and attach or reproduce the observed output below.

```
public class PatientWelcome {

    public static void main(String[] args) {

        String patientName = "Aarav";
        int patientId = 101;
        int age = 21;
        System.out.println("Healthcare Registration");
        System.out.println("----------------------");
        System.out.println("Patient ID   : " + patientId);
        System.out.println("Patient Name : " + patientName);
        System.out.println("Age          : " + age);
        System.out.println("Registration successful.");
    }
}
```

Output -

```
Healthcare Registration
----------------------
Patient ID   : 101
Patient Name : Aarav
Age          : 21
Registration successful.
```

B.3 Observations and Learning

- Identify the role of JVM, JRE and JDK in Java execution.
  JDK provides development tools such as javac and the components needed to develop and run Java programs. JRE provides the libraries and runtime environment needed to run Java applications. JVM loads and executes Java bytecode.
  The same bytecode can run on different operating systems because each operating system can have a compatible JVM that interprets or compiles the bytecode for that platform.
  The main() method is the entry point of a standard Java application. Program execution begins from this method.

- Why can the same Java bytecode run on different operating systems when a compatible JVM is available?
  Java bytecode is platform independent. Different operating systems can use compatible JVM implementations to execute the same bytecode, following Java's Write Once, Run Anywhere (WORA) principle.

- What is the purpose of the main() method?

  The main() method is the entry point of a standard Java application. Program execution begins from this method. The main() method is the entry point of a standard Java application. Program execution begins from this method.

Student observation / learning:

The program was compiled using javac and executed using java. The Java source file (.java) was converted into bytecode (.class), which was executed by the JVM. The program displayed the patient registration details successfully.

B.4 Conclusion

Patient registration is used only as a simple application context consistent with the syllabus theme.

Student conclusion:

The experiment demonstrated the basic structure and execution of a Java program. It helped explain the roles of JDK, JRE, JVM, bytecode, and the main() method. The program successfully displayed the patient registration details.

B.5 Questions of Curiosity

1. Identify the functions of JDK, JRE and JVM.

JDK is used to develop Java programs and includes tools such as the compiler.

JRE provides the environment required to run Java applications.

JVM executes Java bytecode.

2. Why is Java bytecode considered platform independent?
   Java bytecode is platform independent because it can run on different operating systems that have a compatible JVM.

3. What is the role of the main() method?

   The main() method serves as the starting point of a standard Java application. The JVM begins application execution by invoking this method.